Emergency Department Wait Time Prediction: Systematic Literature Review

Final Presentation

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Progress Update

What changed since the last presentation

M2 — Stopped at EDA

- 12 articles in literature review.
- 17 raw feature, EDA only.
- No ML models trained.
- No interpretability layer.
- No fairness analysis.

→

Full ML pipeline + paper

- 12 reviewed studies + 5 method refs.
- 26 features, 5 engineered.
- 5 models trained + 5-fold CV.
- SHAP TreeExplainer applied.
- Fairness audit across 3 dimensions.

Research Questions

RQ1 **Predictive Accuracy & Generalizability**

Can supervised regression models predict ED length of stay in minutes with materially lower error than naive baselines, and does that accuracy generalize across all five ESI triage levels?

RQ2 **Explainability & Demographic Fairness**

Which features actually drive the model's predictions according to SHAP, and does the resulting model produce equitable error across race/ethnicity, gender, and age?

Dataset → Hospital Emergency Room Visits (Kaggle, CC0)

9,216	26	5	0%
ED visits	Features (17 raw + 5 engineered)	ESI triage levels	Missing values after cleaning

Source & Target

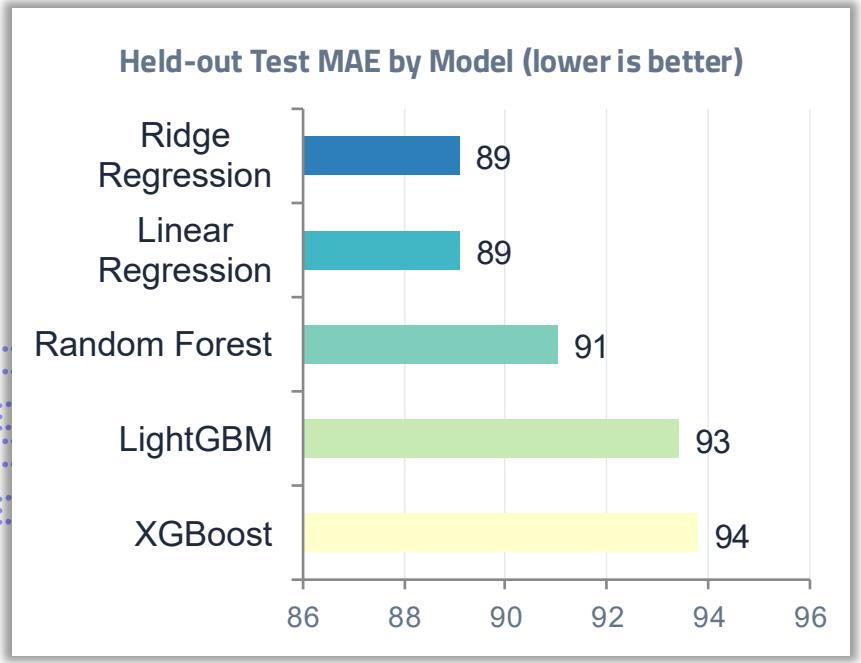
- **Source:** Kaggle (CC0 public domain, no credentialing).
- **Schema mirror:** MIMIC-IV-ED benchmark.
- **Target:** ed_los_minutes (continuous).
- **Distribution:** right-skewed; mean 189.8 min, max 2,644 min.

Feature Groups

- **Triage:** ESI acuity (1-5), pain score.
- **Vitals:** HR, SBP, DBP, temp, O_2, resp rate.
- **Engineered:** Shock index, pulse pressure, age group, arrival shift, is weekend.
- **Demographics:** Age, gender, race/ethnicity (model + fairness audit).

RQ1 - Answer

Yes – but accuracy is not uniform across ESI levels



Best model

Ridge — MAE 89.09 min

Lowest test MAE at 89.09 min and highest test R^2 at 0.141. CV metrics tracked the held-out ranking within ±1 MAE point — ruling out overfitting. Naive baseline sanity floor: 98.71 min.

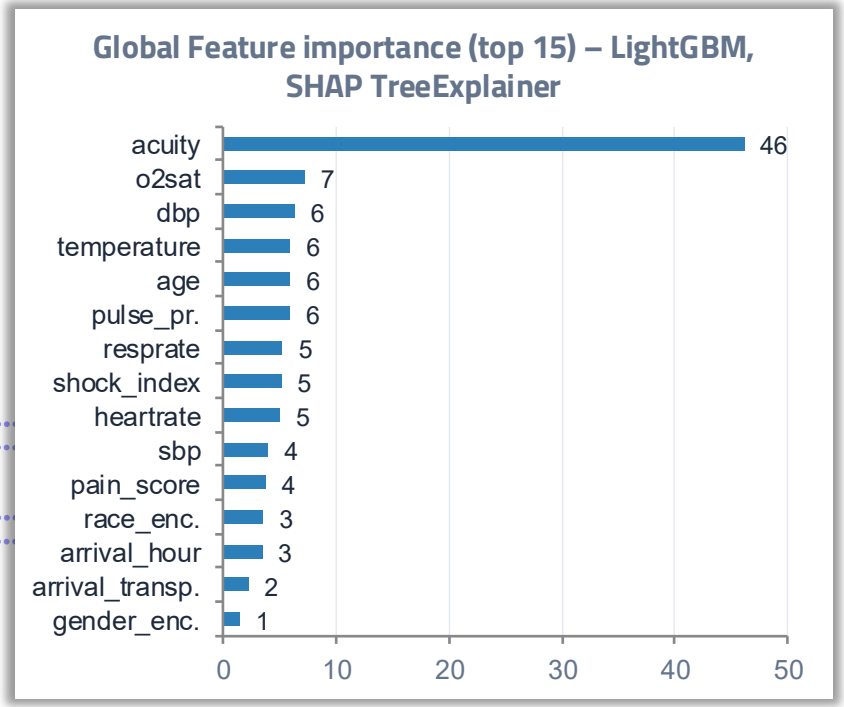
ESI stratification reveals

3.5× MAE spread by acuity

ESI-5 (mean LOS 91 min): MAE 36.96 min. ESI-2 (mean LOS 265 min): MAE 130.28 min. A 3.5× ratio that the single pooled MAE of 89 min would never reveal.

RQ2 - Answer

Acuity dominates; race / age gaps are real



SHAP says

acuity carries ~6.4× the weight

Vital signs (o2sat, dbp, temperature) follow. Demographics, arrival context, and disposition occupy the bottom of the ranking — physiological severity drives LOS more than identity does.

Fairness gaps (MAE in min)

Race / ethnicity	**15.3 min**
Age group	**11.5 min**
Gender	**1.0 min**

→ *subgroup auditing must be a default, not an extension.*

What Makes Our Project Different

Five things together that no prior study delivers

01

Continuous regression

Predicting LOS in minutes — not the binary "prolonged vs. not" used in 10/12 reviewed studies.

02

All five ESI levels

Per-acuity-level evaluation. Most studies report a single pooled metric or only ESI-3.

03

SHAP interpretability

Global feature ranking on LightGBM. Only 4/12 reviewed studies do this rigorously.

04

Demographic fairness

MAE/RMSE disaggregated by race, gender, age. Only 2/12 do any fairness audit.

05

Fully reproducible on public data

CCO dataset, fixed seed, GitHub-published notebook. 9/12 reviewed studies use proprietary single-center data.

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To our knowledge, no prior study delivers all five — on a single, fully public dataset.

Reflections

What went well, what challenged us, what's next

WHAT WORKED

Clean reproducibility

Public CC0 dataset, fixed seed, GitHub-published pipeline. Anyone can re-run the entire study end-to-end in Colab.

CHALLENGES

Counter-finding to defend

Ridge beat XGBoost / LightGBM — opposite of the literature. We resisted tuning around it and explained it as a dataset-size effect.

NEXT STEPS

Three follow-ups

Add NLP on chief-complaint text, replicate on MIMIC-IV-ED for external validity, apply intersectional debiasing.

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